

Literature-Implied Partial Answer to an Additive Lebesgue-Bounds Question

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Status. `literature_implied_answer` (partial Q1 / broad additive-parameter formulation). This is a scoped literature-status packet, not an original proof packet.

Source question. In Section 6, “Further questions,” of Pablo M. Berna, Oscar Blasco, and Gustavo Garrigos, *Lebesgue inequalities for the greedy algorithm in general bases*, arXiv:1606.06721, Q1 asks for bounds for the Lebesgue constants $\mathbf{L}_N$ and $\tilde{\mathbf{L}}_N$ which depend additively on the unconditionality constants and on the democracy/A-property constants. In particular, the source asks when one can have

$$\mathbf{L}_N \lesssim k_N + \nu_N \quad \text{or} \quad \mathbf{L}_N \lesssim k_N + \tilde{\mu}_N.$$

Supporting theorem. Fernando Albiac, Jose L. Ansorena, and Pablo M. Berna, *New parameters and Lebesgue-type estimates in greedy approximation*, arXiv:2104.10912, introduces squeeze-symmetry parameters λ_m and proves in Theorem 1.2, with proof in Section 3, that for every basis in a quasi-Banach space there are constants depending only on the modulus of concavity such that

$$\mathbf{L}_m \simeq \max\{k_m, \lambda_m\}.$$

Equivalently, up to the same structural constants,

$$\mathbf{L}_m \lesssim k_m + \lambda_m \quad \text{and} \quad \max\{k_m, \lambda_m\} \lesssim \mathbf{L}_m.$$

The abstract explicitly frames this as an answer to Temlyakov’s 2011 question asking for natural greedy-type parameters that combine linearly with the unconditionality parameters to determine the growth of the Lebesgue constants.

Identification and limitation. This later theorem gives a clean literature answer to the broad additive-bounds program behind Q1, after replacing ν_N or $\tilde{\mu}_N$ by the squeeze-symmetry parameter λ_N . The implication is agent-identified rather than stated as a direct solution to Q1 of arXiv:1606.06721. It does *not* settle the literal source variants asking whether $k_N + \nu_N$ or $k_N + \tilde{\mu}_N$ suffice, nor does it settle Q2 on arbitrary independent power growth of ν_N and k_N , or Q3 on $\gamma_N \approx 1$ with maximal g_N growth.

Search bounds. On 2026-06-26, exact local index search for arXiv:1606.06721 and the source title found no prior fa_banach_001 packet. Local source and web searches for the exact phrases around “ k_N and ν_N grow independently,” “arbitrary $0 < \alpha, \beta \leq 1$,” and “ $\gamma_N \approx 1$ and g_N as large as possible” did not locate a direct resolution. Later nearby papers on Chebyshev/weak greedy algorithms and A-property constants were checked for overlap but did not appear to settle those remaining source questions.

References.

- P. M. Berna, O. Blasco, and G. Garrigos, *Lebesgue inequalities for the greedy algorithm in general bases*, arXiv:1606.06721.
- F. Albiac, J. L. Ansorena, and P. M. Berna, *New parameters and Lebesgue-type estimates in greedy approximation*, arXiv:2104.10912.